



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 11

Team No.: 17

Project Title: **Linux Process Monitoring and Control System**

Team Members

Roll Number	Student Name	Signature
2520090224	Ch. Shanthi Swaroop	
2520090097	V. Sathvik	
2520090223	Gatik Sharma	

1. Abstract

The **Linux Process Monitoring and Control System** is a lightweight tool used to monitor and manage the processes running on a Linux system in real time. Sometimes, certain programs may use too much CPU, memory, or other system resources, which can slow down or even crash the system. This project helps users identify and control such processes.

The system collects important information such as **Process ID (PID), CPU usage, memory usage, process state, I/O activity, and execution time** by using the Linux **/proc filesystem**. It displays this information through a simple terminal-based interface or web dashboard.

In addition to monitoring, the system allows users to control processes. Users can **change process priority using nice/renice, suspend or resume processes, and terminate unwanted or unresponsive processes using signals such as SIGTERM, SIGKILL, and SIGSTOP**.

The system can also detect processes that exceed predefined CPU or memory limits and generate alerts or take automatic actions. The project can be implemented using **C/C++ or Python with POSIX system functions**.

2. Problem Statement

Develop a Linux-based system to monitor running processes, track their CPU and memory usage, and identify processes that consume excessive system resources. The system will also allow users to control or terminate unresponsive processes to improve system performance and stability.

3. Objectives

(List the specific objectives of the project.)

1. To monitor running processes in real time.
2. To track CPU, memory, and process usage.
3. To detect processes using excessive system resources.
4. To improve overall system performance and stability.

4. Proposed Methodology

The Linux Process Monitoring and Control System will be developed to continuously monitor the processes running on a Linux system. The system will collect process information such as **PID, CPU usage, memory usage, process state, and execution time** using the Linux **/proc filesystem**. The collected information will be displayed to the user through a simple terminal interface or web dashboard.

The system will compare CPU and memory usage with predefined threshold values to identify abnormal or high-resource processes. Users will be able to control selected processes by changing their priority, suspending them, or terminating them using Linux system calls and POSIX signals.

Flow:

Read Process Information → Analyze Resource Usage → Display Process Details → Detect High Resource Usage → Alert User → Control/Suspend/Terminate Process

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Monitoring and controlling running processes.
Filesystem	Reading process information such as CPU and memory usage.
Process ID (PID)	Identifying and managing individual processes.
Process Scheduling and Priority	Using nice() and setpriority() to change process priority.
Process Control	Suspending, resuming, and terminating processes.
System Monitoring	Tracking CPU, memory, and process resource usage.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090097	V. Sathvik	Process Monitoring – Collect process details such as PID, CPU usage, memory usage, and process state using the /proc filesystem.
2520090224	Ch. Shanthi Swaroop	Process Control – Implement process priority management, suspend/resume functionality, and process termination using POSIX signals.
2520090223	Gatik Sharma	User Interface, Testing & Documentation – Develop the interface, implement resource-usage detection, test the system, and prepare project documentation.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Main operating system used for developing and testing the project.
C / C++	Used to develop the process monitoring and control system.
GCC / G++ Compiler	Used to compile the C/C++ program.
/proc Filesystem	Used to collect information about running processes and system resources.
POSIX APIs	Used for process control, priority management, and signals.
VS Code / Terminal	Used for writing, compiling, and running the program.

8. Expected Outcome

The project is expected to provide a **Linux-based Process Monitoring and Control System** that can display information about running processes, including their **PID, CPU usage, memory usage, and process state**.

The system will detect processes that consume excessive resources and allow users to **change their priority, suspend, or terminate them**. This will help users monitor system performance, identify problematic processes, and improve the **stability and efficiency of the Linux system**.

9. GitHub Repository

GitHub Repository Link: https://github.com/Swaroop2108/OS_Project

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: Ms. D. Alekhya

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____